BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 288 · Symmetric and asymmetric encryption

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which technique protects data against casual interception?

Q2. What does a network proxy do?

Q3. What is the difference between confidentiality and integrity?

Q4. Rising-edge D flip-flop Q=0, D=1 just before next edge. What happens to Q, and when?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

symmetric — Shared-secret encryption.

asymmetric — Public/private-key cryptography.

public key — Distributable key in a pair.

private key — Secret member of key pair.

TLS — Protocol establishing secure authenticated channels.

New learning

Understand — Symmetric encryption uses the same secret key for encryption and decryption. The difficulty is sharing that key securely.

Trace — Asymmetric encryption uses a public/private key pair. Public-key encryption for confidentiality uses a recipient's public key to encrypt and their private key to decrypt.

Evaluate — Real secure connections typically use a hybrid protocol: authenticated asymmetric cryptography to establish secrets, then efficient symmetric encryption of application data.

Worked example

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

1. Predict first — Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

2. Show method — Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

We Do · Practise with guidance

Bob posts his public encryption key. Alice wants only Bob to read a message. Which key is used?

```
Encrypt(BobPublic, message)
Decrypt(BobPrivate, ciphertext)
```

Predict first

Alice encrypts to Bob's public key without needing to learn Bob's private key.

Show method

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Bob retains the private key and decrypts. The public key must be authenticated to prevent substitution.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Describe symmetric and asymmetric key arrangements. [6]

Question 2

Explain how a secure website can combine asymmetric and symmetric methods. [6]

Question 3

Evaluate symmetric-only encryption for a school communicating with 200 external partners. [6]

Exit Ticket · Answer before checking

A public key must be kept secret to use public-key encryption.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Encryption.

Q2. Acts as an intermediary for requests/responses.

Q3. Confidentiality prevents unauthorised reading; integrity detects unauthorised changes.

Q4. Q changes to 1 on the next rising edge, not simply when D changes.

You Do pages · Independent questions

Q1. Symmetric encrypt/decrypt use shared secret [2]; asymmetric uses public/private pair [2], encrypt to recipient's public key [1], private key decrypts [1].

Q2. Server identity authenticated via certificate [2]; authenticated key agreement establishes session secret [2]; symmetric session encryption protects bulk traffic efficiently [2].

Q3. Fast for bulk data [1]; each shared secret needs secure exchange [2]; managing many unique shared keys is difficult [1]; asymmetric authenticated exchange/hybrid design improves distribution [2].

Exit Ticket · Model answer

False. The paired private key is secret; the public key needs authentic distribution.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding